PARISHRAM 2.0

CLASS 12TH 2026

Relations and Functions

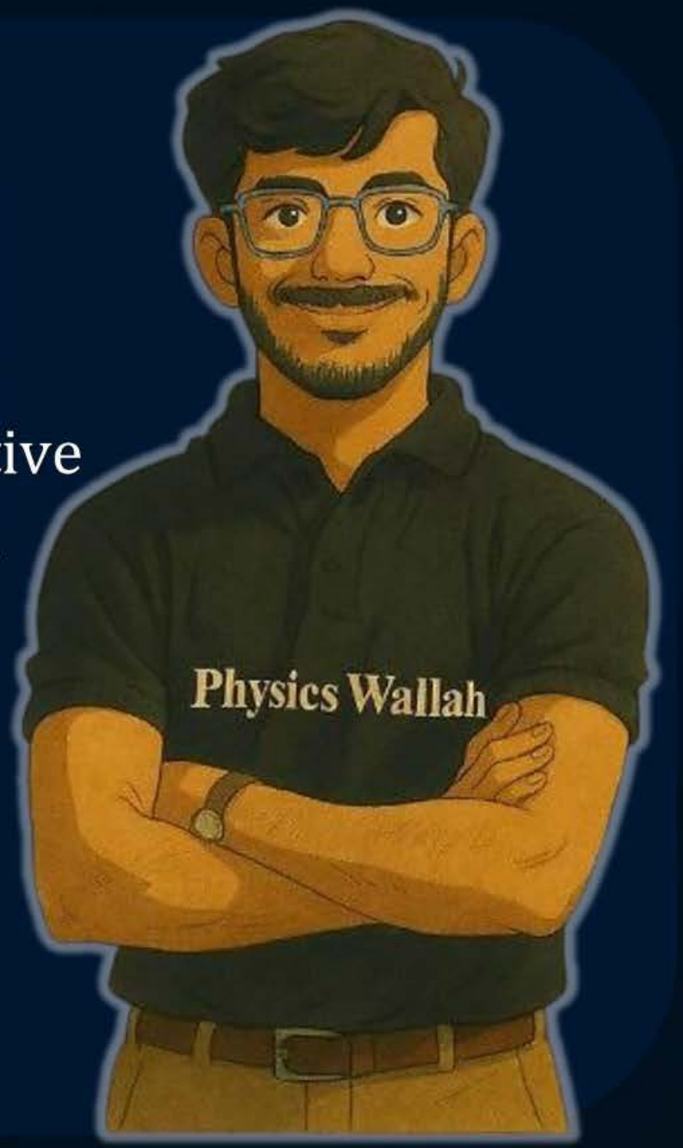
MATHS **LECTURE-3**

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TOPICS TO BE COVERED

1. Recap
2. Questions Based on Reflexive, Symmetric and Transitive



Recap

Reflexive, Symmetric & Transitive

$A = \{1, 2, 3\}$

$R : A \to A$

* Reflexive

$R_1 = \{(1,1)(2,2)(3,3)\}$ ✓

$R_2 = \{(1,1)(2,2)(1,2)(3,3)\}$ ✓

* Symmetric

$R_1 = \{(1,2)(2,1)\}$ ✓

$R_2 = \{(3,2)(1,4)(2,3)\}$ ✗

$R_3 = \{(1,2)(2,1)(3,3)\}$ ✓

If $(a,b) \in R$

$(b,c) \in R$

then $(a,c) \in R$

* Transitive

$R_1 = \{(1,2),(2,3),(1,3)\}$ ✓

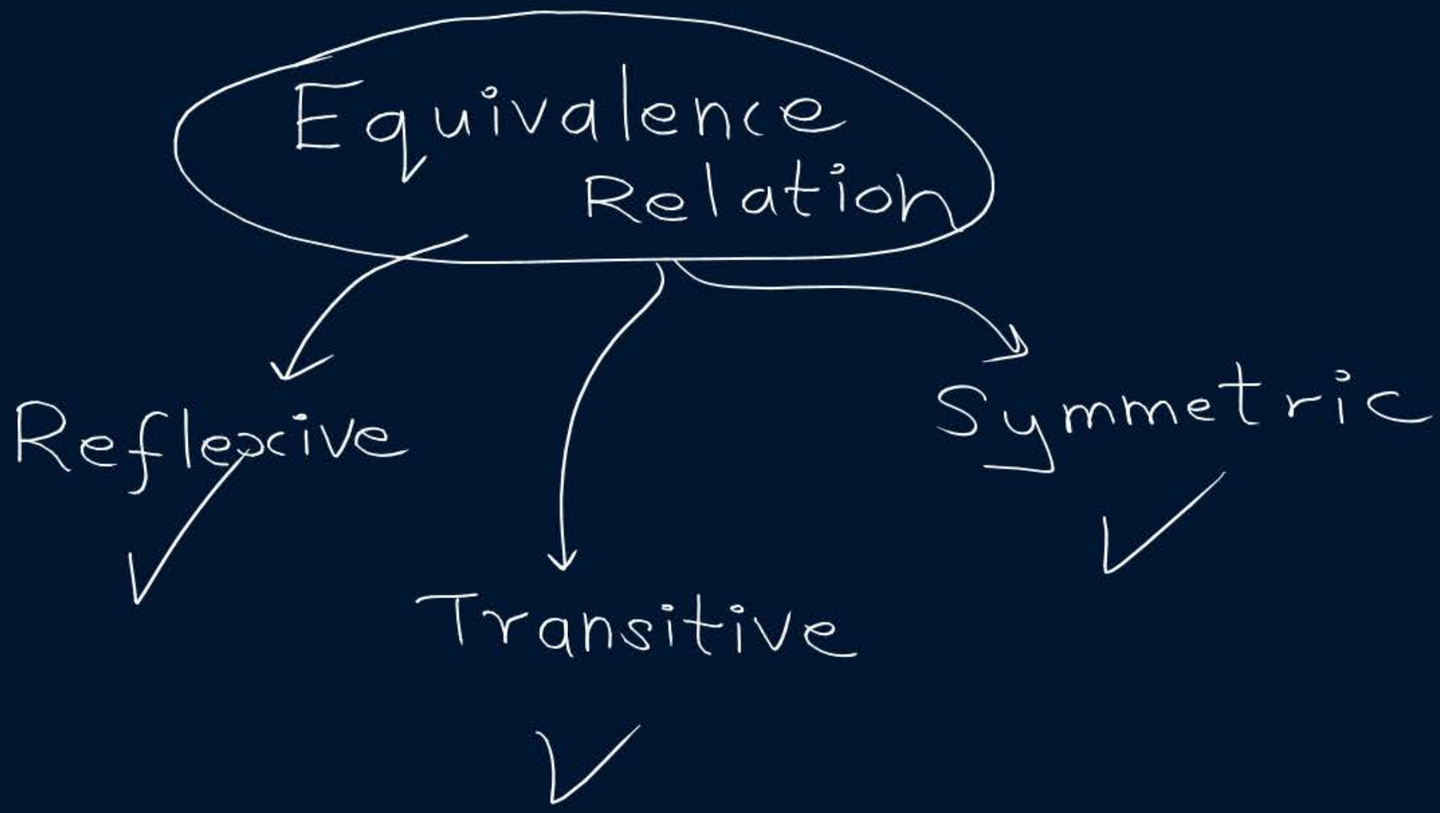
(a b b c a c)

$R_2 = \{(1,2)(2,1)(1,1)\}$ ✓

(a b b c a c)

$R_3 = \{(3,2)\}$ ✓

Equivalence Relation



QUESTION

Determine whether each of the following relations are reflexive, symmetric and transitive:

(iii) Relation R in the set $A = \{1, 2, 3, 4, 5, 6\}$ as
$R = \{(x, y) : y \text{ is divisible by } x\}$

$R = \{(1,1)(1,2)(1,3)(1,4)(1,5)(1,6)$
$(2,2)(2,4)(2,6)(3,3)(3,6)(4,4)(5,5),(6,6)\}$

Reflexive → Sabke Twins?? ✓

Symmetric → Ulta ?? $(1,2) \in R$
$(2,1) \notin R$ ✗

Transitive → $(1,2) \in R$ $(1,4) \in R$ ✓
$(2,4) \in R$

QUESTION

Show that that relation R in the set R of real numbers, defined as $R = \{(a, b) : a \leq b^2\}$ is neither reflexive nor symmetric nor transitive.

$R = \{(a,b) : a \leq b^2\}$

Reflexive

$R = \{(a,a) : a \leq a^2\}$?

$a = 3 \quad 3 \leq 3^2$

$3 \leq 9$ ✓

Not Reflexive ✗

$a = \frac{1}{2}$

$0.5 < 0.25$ ✗

$\frac{1}{2} \leq \left(\frac{1}{2}\right)^2$

$\frac{1}{2} \leq \frac{1}{4}$ ✗

Symmetric

$R \rightarrow (a,b) \rightarrow a \leq b^2$

$(b,a) \quad b \leq a^2$ ✗

$(1,2)$

a, b

$1 \leq 2^2$ ✓

$(2,1) \quad 2 \leq 1^2$ ✗

Not Symmetric

For transitive

$a=3,\ b=-2,\ c=1$

$(a,b)\in R$	$a \le b^2$	$3 \le (-2)^2 \rightarrow 3 \le 4$ ✓
$(b,c)\in R$	$b \le c^2$	$-2 \le (1)^2 \rightarrow -2 \le 1$ ✓
$(a,c)\in R$	$a \le c^2$ → check	$3 \le 1^2 \rightarrow 3 \le 1$ ✗

Transitive ✗

QUESTION

Let T be the set of all triangles in a plane with R a relation in T given by $R = \{(T_1, T_2) : T_1 \text{ is congruent to } T_2\}$. Show that R is an equivalence relation

For Practice

THANK YOU